

# Stresses

Stress at a point

Stress Tensor

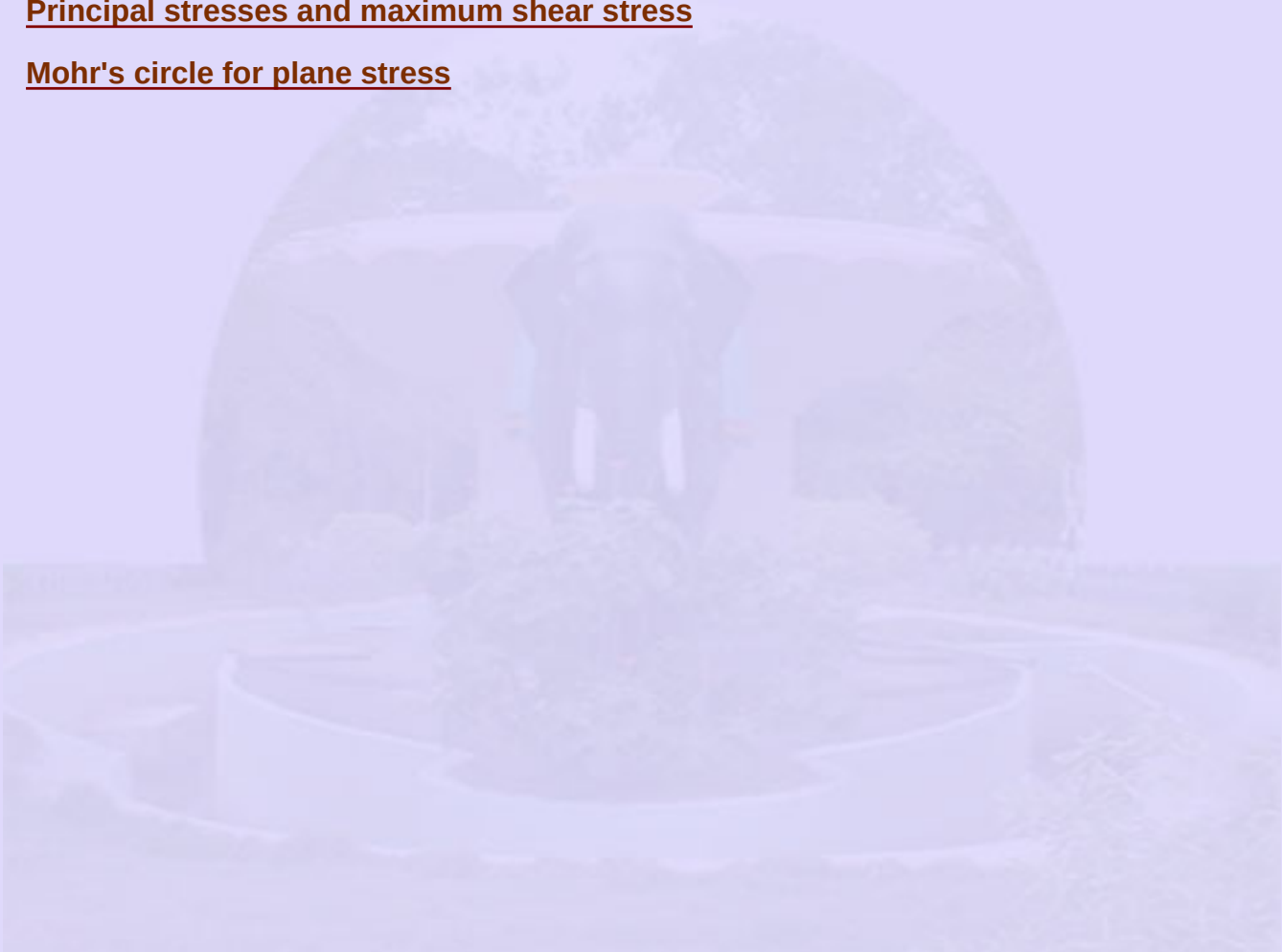
Equations of Equilibrium

Different states of stress

Transformation of plane stress

Principal stresses and maximum shear stress

Mohr's circle for plane stress



## Introduction

### 2.1 stress at a point

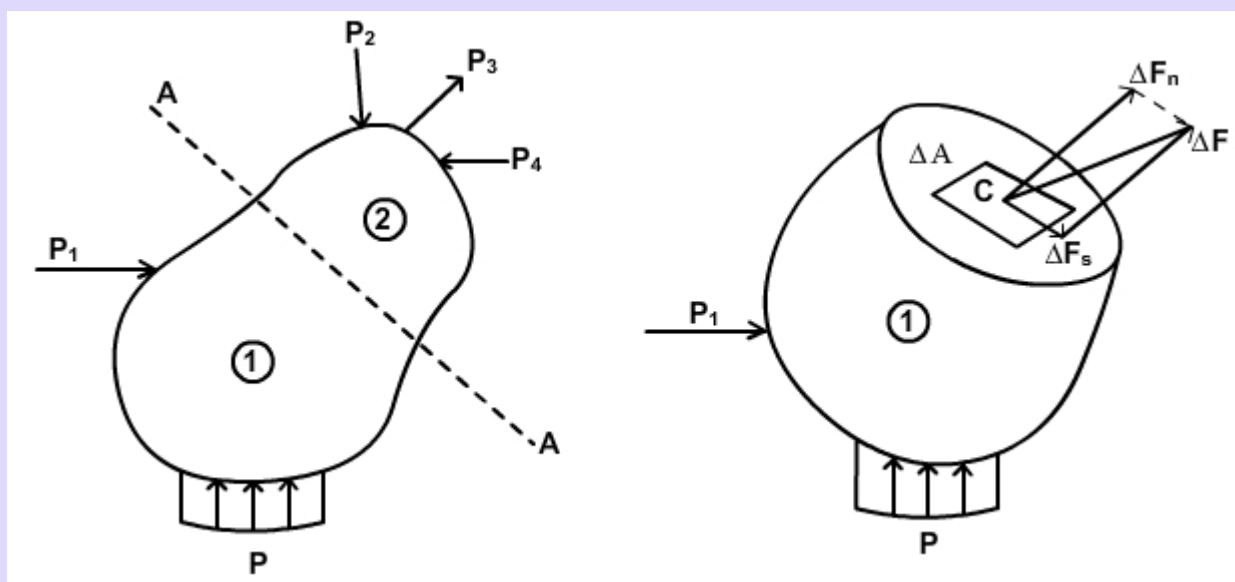


Figure 2.1

Consider a body in equilibrium under point and traction loads as shown in figure 2.1.

After cutting the body along section AA, take an infinitesimal area  $\Delta A$  lying on the surface consisting a point C.

The interaction force between the cut sections 1 & 2, through  $\Delta A$  is  $\Delta F$ . Stress at the point C can be defined,

$$\sigma = \lim_{\Delta A \rightarrow 0} \frac{\Delta F}{\Delta A} \quad 2.1$$

$\Delta F$  is resolved into  $\Delta F_n$  and  $\Delta F_s$  that are acting normal and tangent to  $\Delta A$ .

$$\text{Normal stress, } \sigma_n = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_n}{\Delta A} \quad 2.2$$

$$\text{Shear Stress, } \sigma_s = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_s}{\Delta A} \quad 2.3$$

[Top](#)

## 2.2 stress Tensor

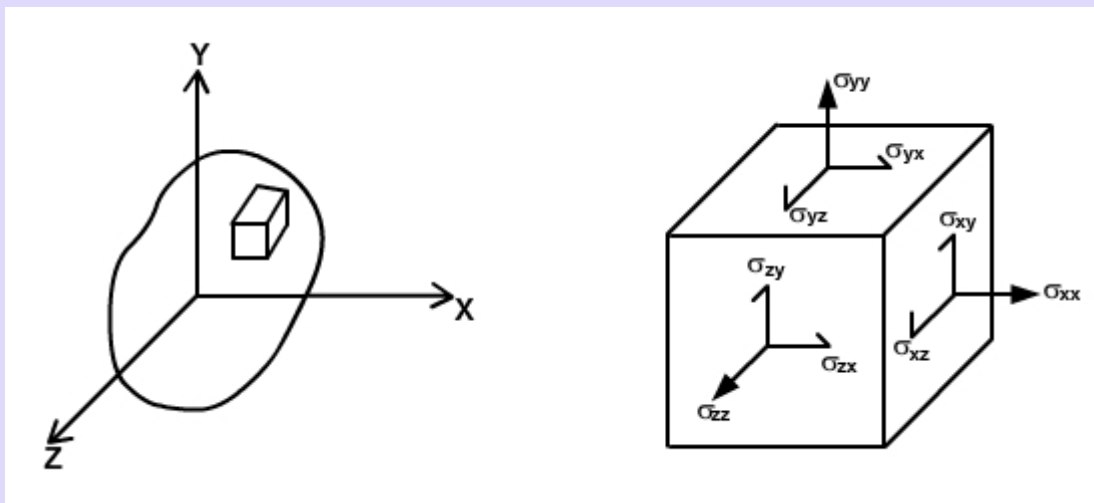


Figure 2.2

Consider the free body diagram of an infinitesimally small cube inside the continuum as shown in figure 2.2.

Stress on an arbitrary plane can be resolved into two shear stress components parallel to the plane and one normal stress component perpendicular to the plane.

Thus, stresses acting on the cube can be represented as a second order tensor with nine components.

$$\sigma = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix} \quad 2.4$$

Is stress tensor symmetric?

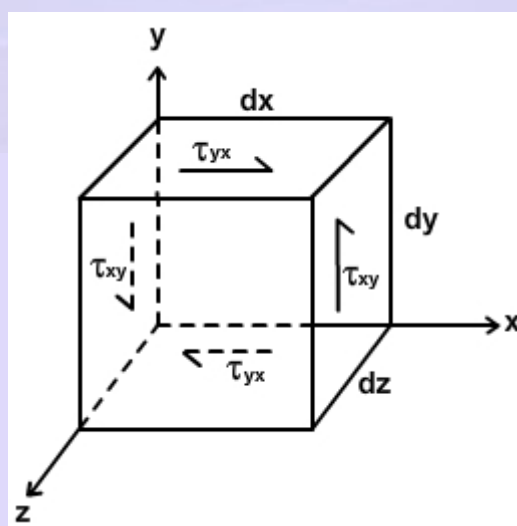


Figure 2.3

Consider a body under equilibrium with simple shear as shown in figure 2.3.

Taking moment about z axis,

$$M_z = (\tau_{yx} d_x d_z) d_y - (\tau_{xy} d_y d_z) d_x = 0$$

$$\boxed{\tau_{xy} = \tau_{yx}}$$

Similarly,  $\tau_{xz} = \tau_{zx}$  and  $\tau_{yz} = \tau_{zy}$ .

Hence, the stress tensor is symmetric and it can be represented with six components,  $\sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \tau_{xy}, \tau_{xz}$  and  $\tau_{yz}$ , instead of nine components.

$$\sigma = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{bmatrix} \quad 2.5$$

**Top**

## 2.3 Equations of Equilibrium

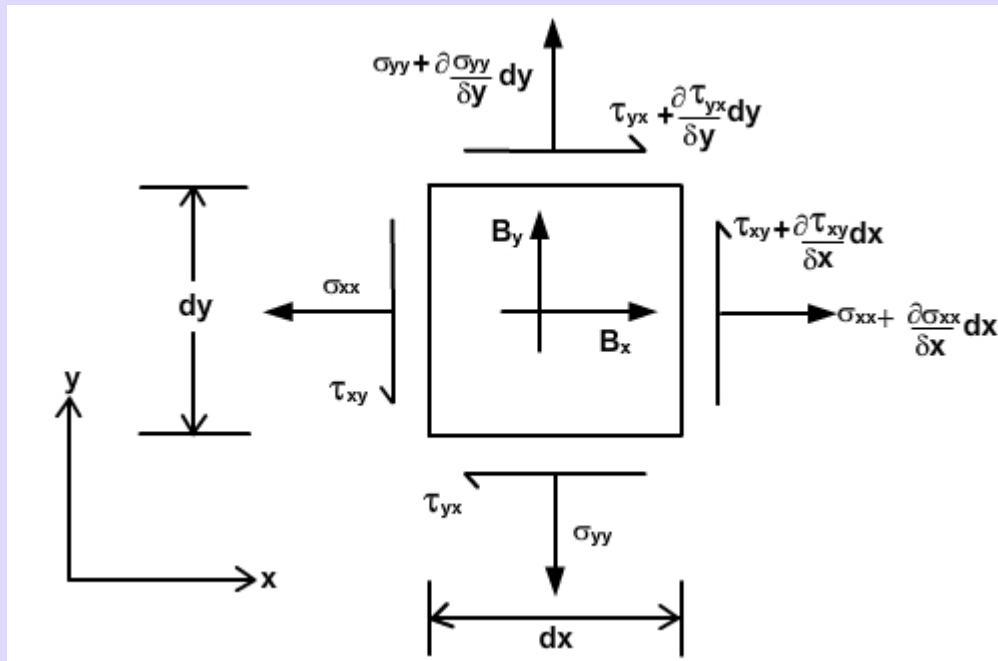


Figure 2.4

Consider an infinitesimal element of a body under equilibrium with sides  $dx \times dy \times 1$  as shown in figure 2.4.

$B_x$ ,  $B_y$  are the body forces like gravitational, inertia, magnetic, etc., acting on the element through its centre of gravity.

$$\sum F_x = 0,$$

$$\left( \sigma_{xx} + \frac{\partial \sigma_{xx}}{\partial x} dx \right) (dy \times 1) - \sigma_{xx} (dy \times 1) + \left( \tau_{yx} + \frac{\partial \tau_{yx}}{\partial y} dy \right) (dx \times 1) - \tau_{yx} (dx \times 1) + B_x (dx \times dy \times 1) = 0$$

Similarly taking  $\sum F_y = 0$  and simplifying, equilibrium equations of the element in differential form are obtained as,

$$\begin{aligned} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + B_x &= 0 \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + B_y &= 0 \end{aligned} \tag{2.6}$$

Extending this derivation to a three dimensional case, the differential equations of equilibrium become,

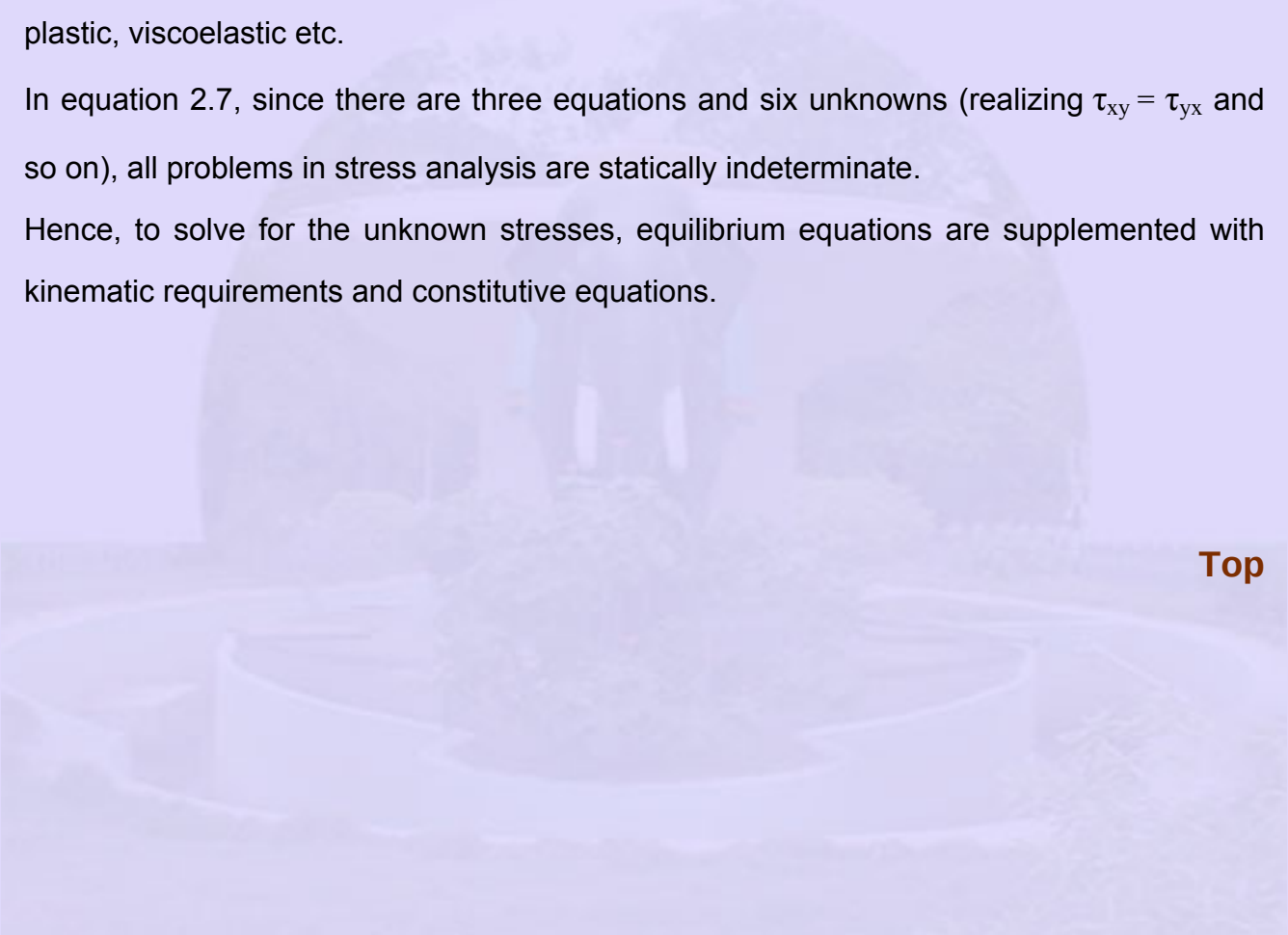
$$\begin{aligned}
 \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + B_x &= 0 \\
 \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + B_y &= 0 \\
 \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + B_z &= 0
 \end{aligned}
 \tag{2.7}$$

When the right hand side of above equations is not equal to zero, then, they become equations of motion.

Equations of equilibrium are valid regardless of the materials nature whether elastic, plastic, viscoelastic etc.

In equation 2.7, since there are three equations and six unknowns (realizing  $\tau_{xy} = \tau_{yx}$  and so on), all problems in stress analysis are statically indeterminate.

Hence, to solve for the unknown stresses, equilibrium equations are supplemented with kinematic requirements and constitutive equations.

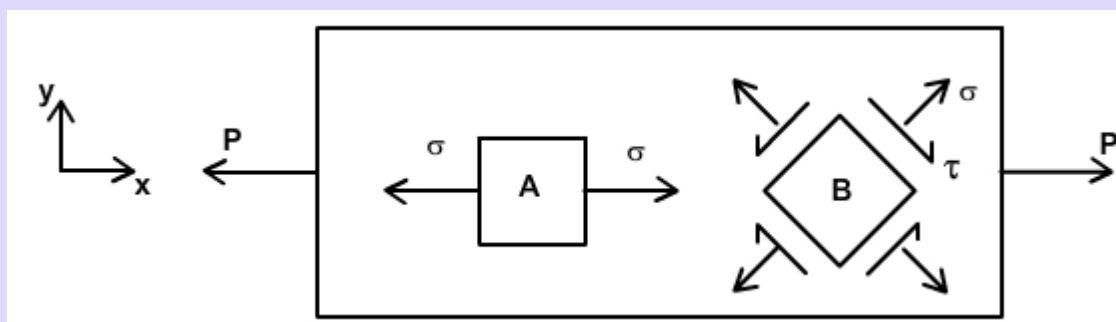


**Top**

## 2.4 Different states of stress

Depending upon the state of stress at a point, we can classify it as uniaxial(1D), biaxial(2D) and triaxial(3D) stress.

### 2.4.1 One dimensional stress(Uniaxial)



**Figure 2.5**

Consider a bar under a tensile load  $P$  acting along its axis as shown in figure 2.5.

Take an element  $A$  which has its sides parallel to the surfaces of the bar.

It is clear that the element has only normal stress along only one direction, i.e.,  $x$  axis and all other stresses are zero. Hence it is said to be under uni-axial stress state.

Now consider another element  $B$  in the same bar, which has its slides inclined to the surfaces of the bar.

Though the element has normal and shear stresses on each face, it can be transformed into a uni-axial stress state like element  $A$  by transformation of stresses (will be discussed in section 2.5).

Hence, if the stress components at a point can be transformed into a single normal stress (principal stress as will be discussed later), then, the element is under uni-axial stress state.

Is an element under pure shear uni-axial?

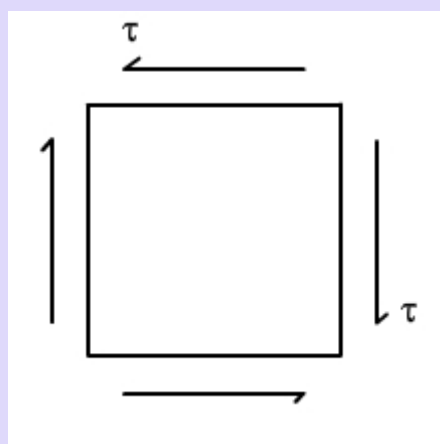


Figure 2.6

The given stress components cannot be transformed into a single normal stress along an axis but along two axes. Hence this element is under biaxial / two dimensional stress state.

#### **2.4.2 Two dimensional stress (Plane stress)**

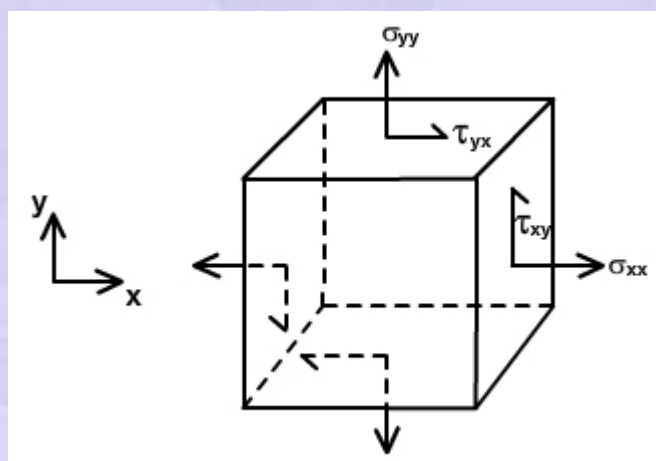


Figure 2.7

When the cubic element is free from any of the stresses on its two parallel surfaces and the stress components in the element can not be reduced to a uni-axial stress by transformation, then, the element is said to be in two dimensional stress/plane stress state. Thin plates under mid plane loads and the free surface of structural elements may experience plane stresses as shown below.



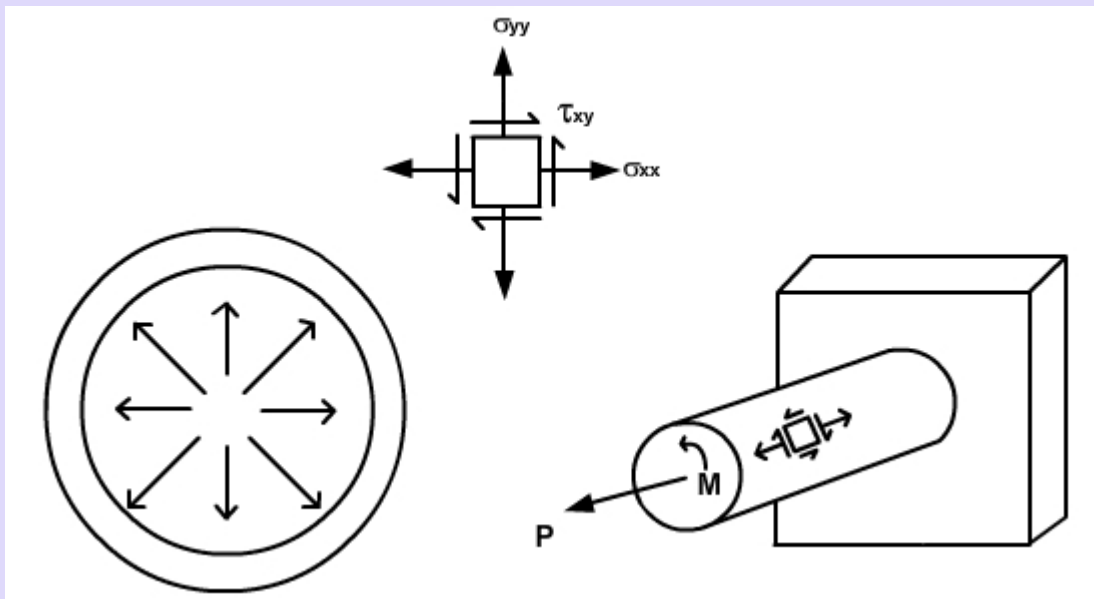


Figure 2.8

[Top](#)

## 2.5 Transformation of plane stress

Though the state of stress at a point in a stressed body remains the same, the normal and shear stress components vary as the orientation of plane through that point changes.

Under complex loading, a structural member may experience larger stresses on inclined planes than on the cross section.

The knowledge of maximum normal and shear stresses and their plane's orientation assumes significance from failure point of view.

Hence, it is important to know how to transform the stress components from one set of coordinate axes to another set of co-ordinates axes that will contain the stresses of interest.

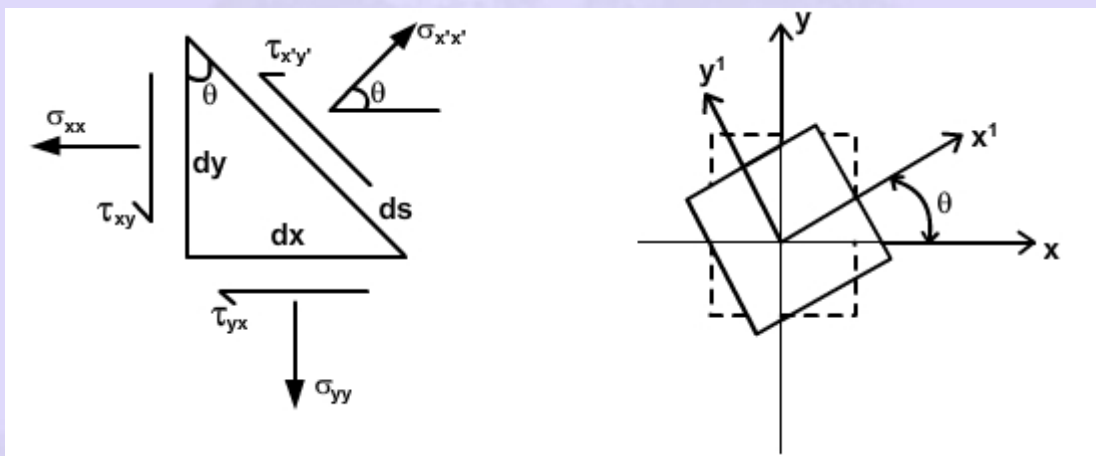


Figure 2.9

Consider a prismatic element with sides  $dx$ ,  $dy$  and  $ds$  with their faces perpendicular to  $y$ ,  $x$  and  $x'$  axes respectively. Thickness of the element is  $t$ .

$\sigma_{x'x'}$  and  $\tau_{x'y'}$  are the normal and shear stresses acting on a plane inclined at an angle  $\theta$  measured counter clockwise from  $x$  plane.

Under equilibrium,  $\sum F_{x'} = 0$

$$\sigma_{x'x'} \cdot t \cdot ds - \sigma_{xx} \cdot t \cdot dy \cdot \cos \theta - \sigma_{yy} \cdot t \cdot dx \cdot \sin \theta - \tau_{xy} \cdot t \cdot dy \cdot \sin \theta - \tau_{yx} \cdot t \cdot dx \cdot \cos \theta = 0$$

Dividing above equation by  $t \cdot ds$  and using  $\frac{dy}{ds} = \cos \theta$  and  $\frac{dx}{ds} = \sin \theta$ ,

$$\sigma_{x'x'} = \sigma_{xx} \cos^2 \theta + \sigma_{yy} \sin^2 \theta + 2\tau_{xy} \sin \theta \cos \theta$$

Similarly, from  $\sum F_{y'} = 0$  and simplifying,

$$\tau_{x'y'} = (\sigma_{yy} - \sigma_{xx}) \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta)$$

Using trigonometric relations and simplifying,

$$\begin{aligned}\sigma_{x'x'} &= \frac{\sigma_{xx} + \sigma_{yy}}{2} + \frac{\sigma_{xx} - \sigma_{yy}}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \\ \tau_{x'y'} &= \frac{-(\sigma_{xx} - \sigma_{yy})}{2} \sin 2\theta + \tau_{xy} \cos 2\theta\end{aligned}\tag{2.8}$$

Replacing  $\theta$  by  $\theta + 90^\circ$ , in  $\sigma_{x'x'}$  expression of equation 2.8, we get the normal stress along  $y'$  direction.

$$\sigma_{y'y'} = \frac{\sigma_{xx} + \sigma_{yy}}{2} - \frac{\sigma_{xx} - \sigma_{yy}}{2} \cos 2\theta - \tau_{xy} \sin 2\theta\tag{2.9}$$

Equations 2.8 and 2.9 are the transformation equations for plane stress using which the stress components on any plane passing through the point can be determined.

Notice here that,

$$\sigma_{xx} + \sigma_{yy} = \sigma_{x'x'} + \sigma_{y'y'}\tag{2.10}$$

Invariably, the sum of the normal stresses on any two mutually perpendicular planes at a point has the same value. This sum is a function of the stress at that point and not on the orientation of axes. Hence, this quantity is called *stress invariant* at that a point



[Top](#)

## 2.6 Principal stresses and maximum shear stress

From transformation equations, it is clear that the normal and shear stresses vary continuously with the orientation of planes through the point.

Among those varying stresses, finding the maximum and minimum values and the corresponding planes are important from the design considerations.

By taking the derivative of  $\sigma_{x'x'}$  in equation 2.8 with respect to  $\theta$  and equating it to zero,

$$\frac{d\sigma_{x'x'}}{d\theta} = -(\sigma_{xx} - \sigma_{yy})\sin 2\theta + 2\tau_{xy}\cos 2\theta = 0$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_{xx} - \sigma_{yy}} \quad 2.11$$

Here,  $\theta_p$  has two values  $\theta_{p1}$ , and  $\theta_{p2}$  that differ by  $90^\circ$  with one value between  $0^\circ$  and  $90^\circ$  and the other between  $90^\circ$  and  $180^\circ$ .

These two values define the *principal planes* that contain maximum and minimum stresses.

Substituting these two  $\theta_p$  values in equation 2.8, the maximum and minimum stresses, also called as *principal stresses*, are obtained.

$$\sigma_{\max,\min} = \frac{\sigma_{xx} + \sigma_{yy}}{2} \pm \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2} \quad 2.12$$

The plus and minus signs in the second term of equation 2.12, indicate the algebraically larger and smaller principal stresses, i.e. maximum and minimum principal stresses.

In the second equation of 2.8, if  $\tau_{x'y'}$  is taken as zero, then the resulting equation is same as equation 2.11.

Thus, the following important observation pertained to principal planes is made.

The shear stresses are zero on the principal planes

To get the maximum value of the shear stress, the derivative of  $\tau_{x'y'}$  in equation 2.8 with respect to  $\theta$  is equated to zero.

$$\frac{d\tau_{x'y'}}{d\theta} = -(\sigma_{xx} - \sigma_{yy})\cos 2\theta - 2\tau_{xy}\sin 2\theta = 0$$

$$\tan 2\theta_s = \frac{-(\sigma_{xx} - \sigma_{yy})}{2\tau_{xy}} \quad 2.13$$

Hence,  $\theta_s$  has two values,  $\theta_{s1}$  and  $\theta_{s2}$  that differ by  $90^\circ$  with one value between  $0^\circ$  and  $90^\circ$  and the other between  $90^\circ$  and  $180^\circ$ .

Hence, the maximum shear stresses that occur on those two mutually perpendicular planes are equal in algebraic value and are different only in sign due to its complementary property.

Comparing equations 2.11 and 2.13,

$$\tan 2\theta_p = -\frac{1}{\tan 2\theta_s} \quad 2.14$$

It is understood from equation 2.14 that the tangent of the angles  $2\theta_p$  and  $2\theta_s$  are negative reciprocals of each other and hence, they are separated by  $90^\circ$ .

Hence, we can conclude that  $\theta_p$  and  $\theta_s$  differ by  $45^\circ$ , i.e., the maximum shear stress planes can be obtained by rotating the principal plane by  $45^\circ$  in either direction.

A typical example of this concept is given in figure 2.10.

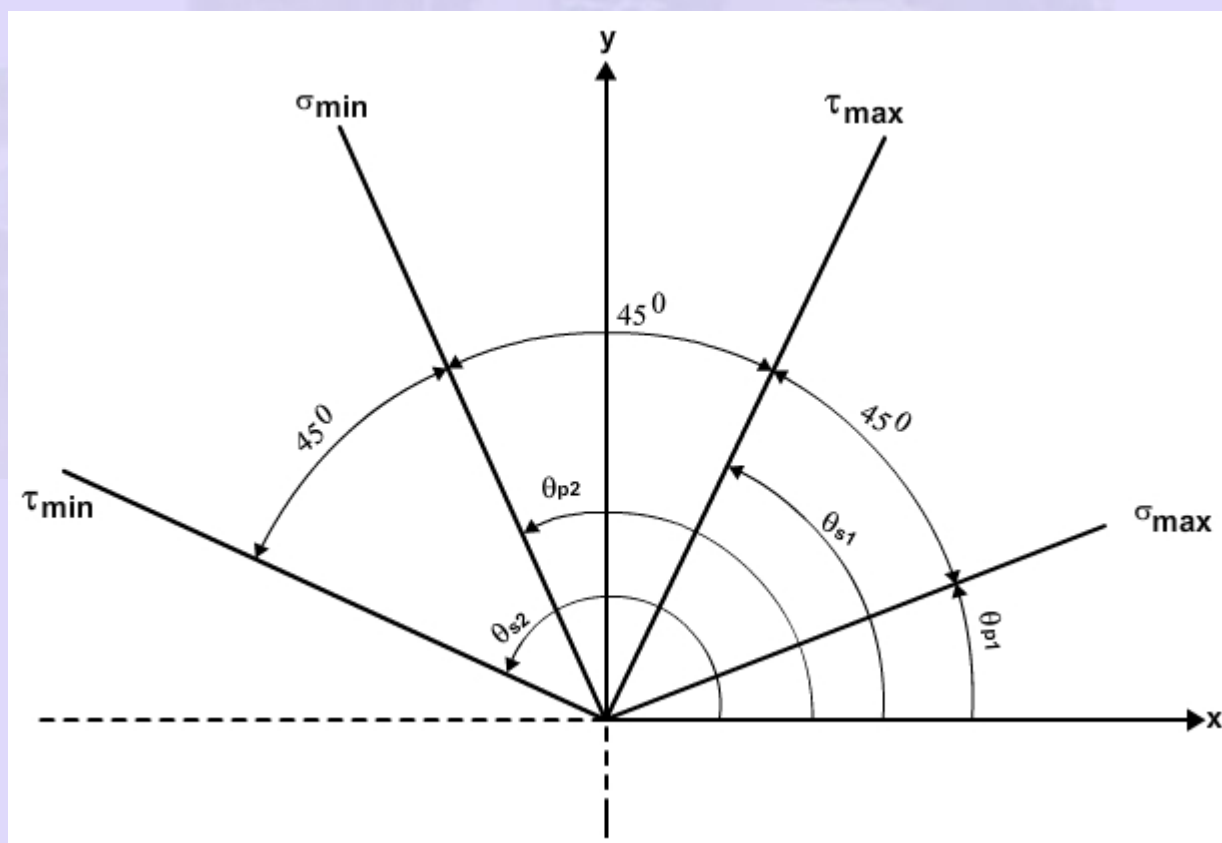


Figure 2.10

The principal planes do not contain any shear stress on them, but the maximum shear stress planes may or may not contain normal stresses as the case may be.

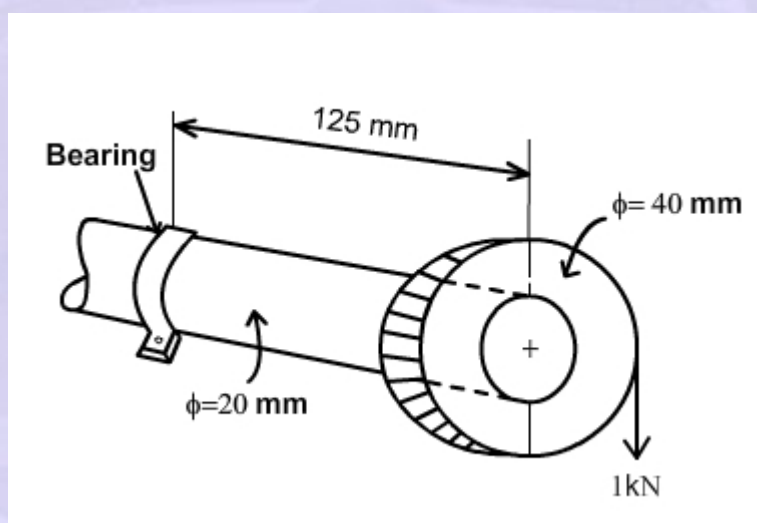
Maximum shear stress value is found out by substituting  $\theta_s$  values in equation 2.8.

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2} \quad 2.15$$

Another expression for  $\tau_{\max}$  is obtained from the principal stresses,

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} \quad 2.16$$

**Example: (Knowledge in torsion and bending is necessary)**

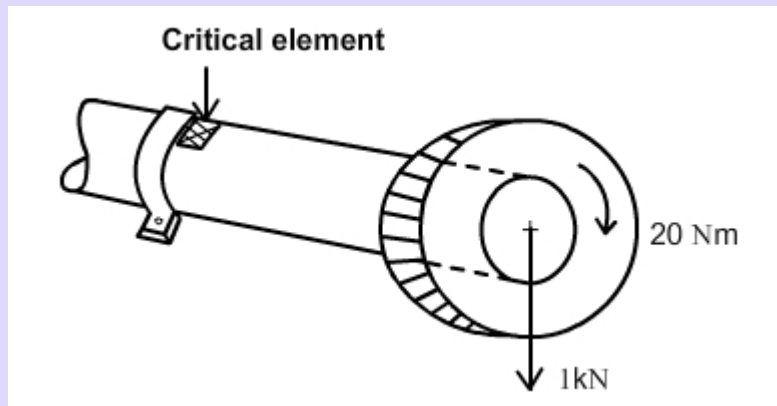


**Figure 2.11**

A gear with a shaft is used to transmit the power as shown in figure 2.11. The load at the gear tooth is 1kN. The diameter of the gear and the shaft are 40 mm and 20 mm respectively. Find the principal stresses and the maximum shear stress on an element, which you feel important on the shaft.

**Solution:**

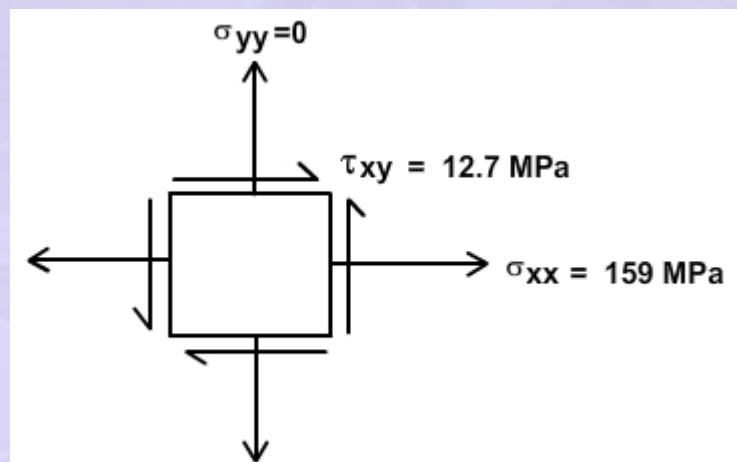
The critical element of design interest lies on the top of the shaft, near to the bearing.



Transfer the load from the end of the gear to the centre of the shaft as a force-couple system with 1kN force and 20 N-m couple clockwise.

$$\text{Bending stress, } \sigma_{\max} = \frac{M}{I} y = \frac{(1 \times 10^3) \times (0.125)}{\pi (0.02)^4 / 64} = 159 \text{ MPa}$$

$$\text{Shearing stress, } \tau_{xy} = \frac{T}{J} r = \frac{20 \times (0.01)}{\pi (0.02)^4 / 32} = 12.7 \text{ MPa}$$



From equation 2.11, the principal plane,

$$\tan 2\theta_p = \frac{2 \times 12.7}{(159 - 0)}$$

$$\theta_{p1} = 4.5^\circ; \theta_{p2} = 90 + 4.5^\circ = 94.5^\circ$$

Using equation 2.8, the principal stresses,

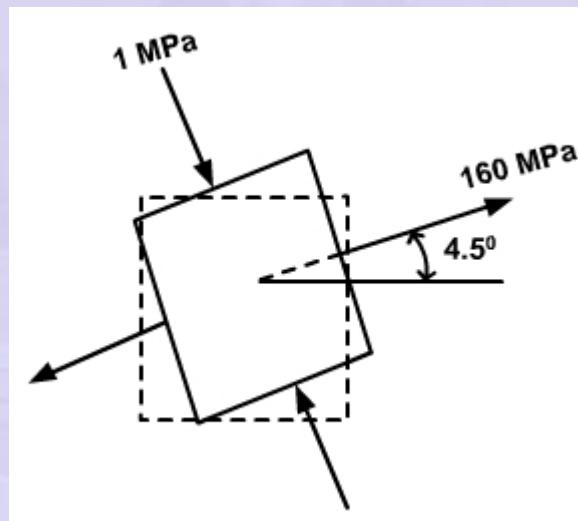
$$\begin{aligned}\sigma_{\max} &= \frac{(159+0)}{2} + \frac{(159-0)}{2} \cos(2 \times 4.5^\circ) + 12.7 \sin(2 \times 4.5^\circ) \\ &= 160 \text{ MPa} \\ \sigma_{\min} &= \frac{(159+0)}{2} + \frac{(159-0)}{2} \cos(2 \times 94.5^\circ) + 12.7 \sin(2 \times 94.5^\circ) \\ &= -1 \text{ MPa}\end{aligned}$$

Alternatively, using equation 2.12,

$$\begin{aligned}\sigma_{\max} &= \left( \frac{159+0}{2} \right) + \sqrt{\left( \frac{159-0}{2} \right)^2 + (12.7)^2} \\ &= 160 \text{ MPa} \\ \sigma_{\min} &= -1 \text{ MPa}\end{aligned}$$

Maximum Shear Stress,  $\tau_{\max} = \frac{160+1}{2} = 80.5 \text{ MPa}$

The principal planes and the stresses acting on them are shown below.



**Top**



## 2.7 Mohr's circle for plane stress

The transformation equations of plane stress 2.8 can be represented in a graphical form which is popularly known as *Mohr's circle*.

Though the transformation equations are sufficient to get the normal and shear stresses on any plane at a point, with Mohr's circle one can easily visualize their variation with respect to plane orientation  $\theta$ .

Besides stress plots, Mohr's circles are used to plot strains, moment of inertia, etc., which follow the same transformation laws as do stresses.

### 2.7.1 Equations of Mohr's circle

Recalling transformation equations 2.8 and rearranging the terms

$$\begin{aligned}\sigma_{x'x'} - \left( \frac{\sigma_{xx} + \sigma_{yy}}{2} \right) &= \frac{\sigma_{xx} - \sigma_{yy}}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \\ \tau_{x'y'} &= - \left( \frac{\sigma_{xx} - \sigma_{yy}}{2} \right) \sin 2\theta + \tau_{xy} \cos 2\theta\end{aligned}\tag{2.17}$$

A little consideration will show that the above two equations are the equations of a circle with  $\sigma_{x'x'}$  and  $\tau_{x'y'}$  as its coordinates and  $2\theta$  as its parameter.

If the parameter  $2\theta$  is eliminated from the equations, then the significance of them will become clear.

Squaring and adding equations 2.17 results in,

$$\left[ \sigma_{x'x'} - \left( \frac{\sigma_{xx} + \sigma_{yy}}{2} \right) \right]^2 + \tau_{x'y'}^2 = \left[ \frac{\sigma_{xx} - \sigma_{yy}}{2} \right]^2 + \tau_{xy}^2\tag{2.18}$$

For simple representation of above equation, the following notations are used.

$$\sigma_{ave} = \frac{\sigma_{xx} + \sigma_{yy}}{2}; \quad r = \left[ \left( \frac{\sigma_{xx} - \sigma_{yy}}{2} \right)^2 + \tau_{xy}^2 \right]^{1/2}\tag{2.19}$$

Equation 2.18 can thus be simplified as,

$$(\sigma_{x'x'} - \sigma_{ave})^2 + \tau_{x'y'}^2 = r^2\tag{2.20}$$

Equation 2.20 represents the equation of a circle in a standard form. This circle has  $\sigma_{x'x'}$  as its abscissa and  $\tau_{x'y'}$  as its ordinate with radius  $r$ .

The coordinate for the centre of the circle is  $\sigma_{x'x'} = \sigma_{ave}$  and  $\tau_{x'y'} = 0$ .

### 2.7.2 Construction procedure

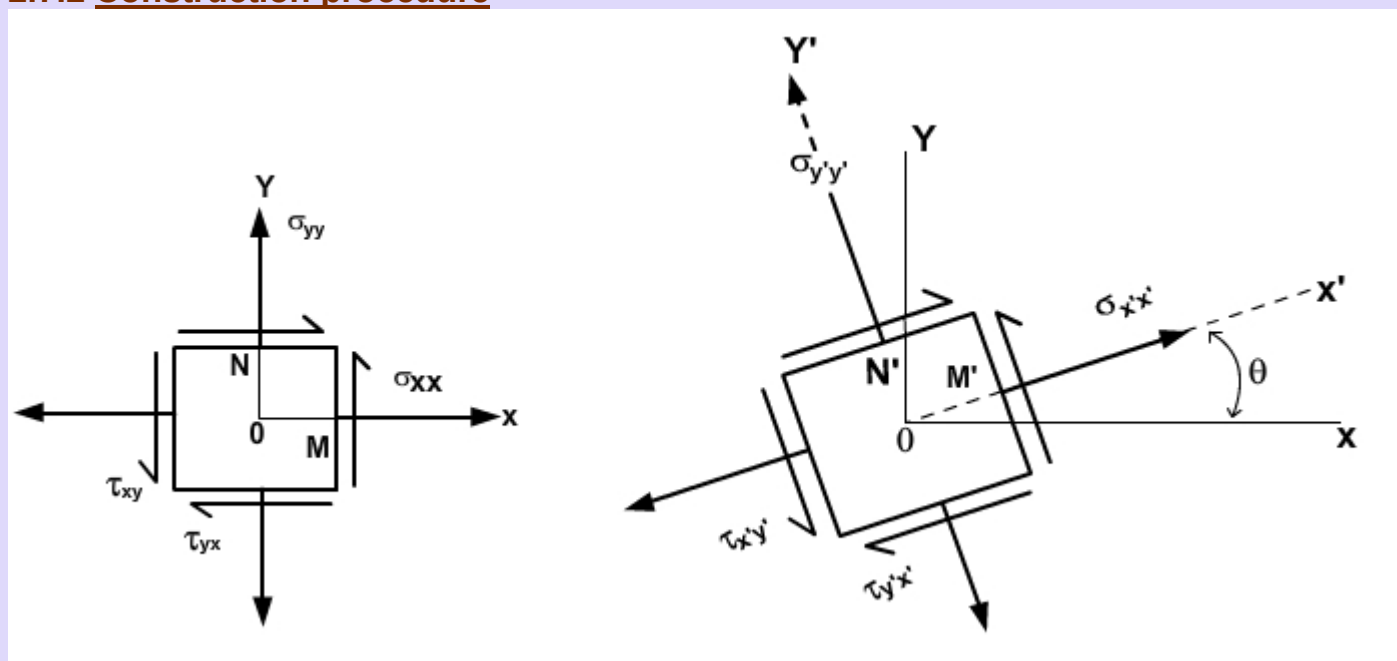


Figure 2.12

Sign convention: Tension is positive and compression is negative. Shear stresses causing clockwise moment about O are positive and counterclockwise negative.

Hence,  $\tau_{xy}$  is negative and  $\tau_{yx}$  is positive.

Mohr's circle is drawn with the stress coordinates  $\sigma_{xx}$  as its abscissa and  $\tau_{xy}$  as its ordinate, and this plane is called the stress plane.

The plane on the element in the material with  $xy$  coordinates is called the physical plane.

Stresses on the physical plane M is represented by the point M on the stress plane with  $\sigma_{xx}$  and  $\tau_{xy}$  coordinates.

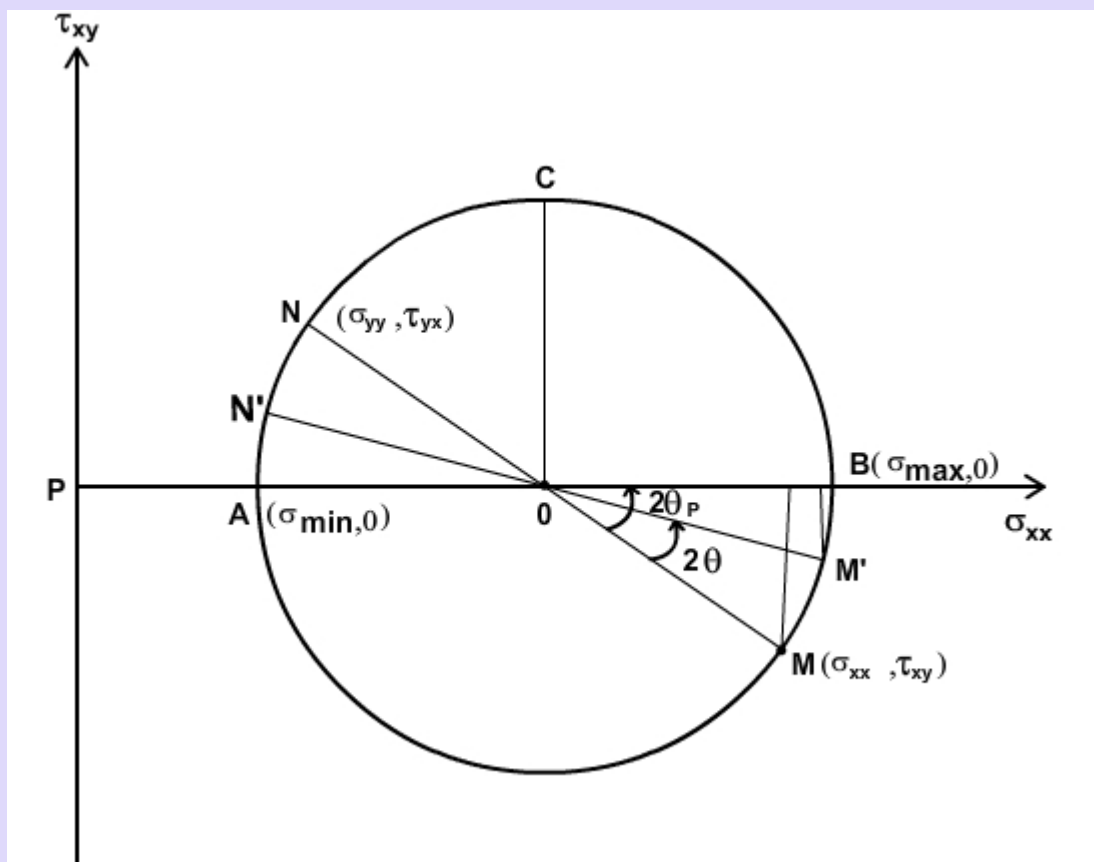


Figure 2.13

Stresses on the physical plane N, which is normal to M, is represented by the point N on the stress plane with  $\sigma_{yy}$  and  $\tau_{yx}$ .

The intersecting point of line MN with abscissa is taken as O, which turns out to be the centre of circle with radius OM.

Now, the stresses on a plane which makes  $\theta^0$  inclination with x axis in physical plane can be determined as follows. Let that plane be M'.

An important point to be noted here is that a plane which has a  $\theta^0$  inclination in physical plane will make  $2\theta^0$  inclination in stress plane.

Hence, rotate the line OM in stress plane by  $2\theta^0$  counterclockwise to obtain the plane M'.

The coordinates of M' in stress plane define the stresses acting on plane M' in physical plane and it can be easily verified.

$$\begin{aligned}\sigma_{x'x'} &= PO + r \cos(2\theta_p - 2\theta) \\ \sigma_{x'x'} &= PO + r [\cos 2\theta_p \cos 2\theta + \sin 2\theta_p \sin 2\theta]\end{aligned}\tag{2.21}$$

Where  $PO = \frac{\sigma_{xx} + \sigma_{yy}}{2}$

$$r = \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}$$

$$\cos 2\theta_p = \frac{\sigma_{xx} - \sigma_{yy}}{2r}$$

$$\sin 2\theta_p = \frac{\tau_{xy}}{2r}$$

Rewriting equation 2.21,

$$\sigma_{x'x'} = \frac{\sigma_{xx} + \sigma_{yy}}{2} + \frac{\sigma_{xx} - \sigma_{yy}}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \quad 2.22$$

Equation 2.22 is same as equation 2.8.

This way it can be proved for shear stress  $\tau_{x'y'}$  on plane M' (do it yourself).

Extension of line M'O will get the point N' on the circle which coordinate gives the stresses on physical plane N' that is normal to M'.

This way the normal and shear stresses acting on any plane in the material can be obtained using Mohr's circle.

Points A and B on Mohr's circle do not have any shear components and hence, they represent the principal stresses,

$$\sigma_{\max, \min} = PO \pm r = \frac{\sigma_{xx} + \sigma_{yy}}{2} \pm \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}$$

The principal plane orientations can be obtained in Mohr's circle by rotating the line OM by  $2\theta_p$  and  $2\theta_p + 180^\circ$  clockwise or counterclockwise as the case may be (here it is counter clock wise) in order to make that line be aligned to axis  $\tau_{xy} = 0$ .

These principal planes on the physical plane are obtained by rotating the plane m, which is normal to x axis, by  $\theta_p$  and  $\theta_p + 90^\circ$  in the same direction as was done in stress plane.

The maximum shear stress is defined by OC in Mohr's circle,

$$\tau_{\max} = \pm r = \pm \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}$$

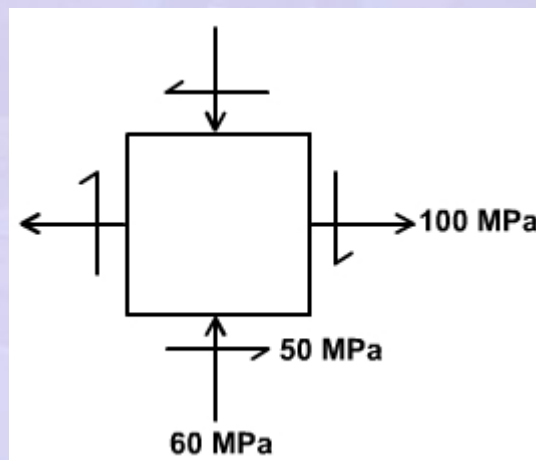
(or)

$$\tau_{\max} = \pm \left(\frac{\sigma_{\max} - \sigma_{\min}}{2}\right)$$

It is important to note the difference in the sign convention of shear stress between analytical and graphical methods, i.e., physical plane and stress plane. The shear stresses that will rotate the element in counterclockwise direction are considered positive in the analytical method and negative in the graphical method. It is assumed this way in order to make the rotation of the elements consistent in both the methods.

### Example:

For the state of plane stress given, determine the principal planes, the principal stresses and the maximum shear stress. Also find the stress components on the element after it is rotated by  $20^\circ$  counterclockwise.



### Solution:

#### Analytical solution:

$$\begin{aligned}\sigma_{\max, \min} &= \frac{100 + (-60)}{2} \pm \sqrt{\left[\frac{100 - (-60)}{2}\right]^2 + (-50)^2} \\ &= 20 \pm 94.34 \text{ MPa}\end{aligned}$$

Maximum principal stress = 114.34 MPa

Minimum principal stress = -74.4 MPa

Principal planes,  $\tan 2\theta_p = \frac{2 \times (-50)}{100 - (-60)}$

$$\theta_{P1} = -16^\circ$$

$$\theta_{P2} = -16^\circ + 90^\circ = 74^\circ$$

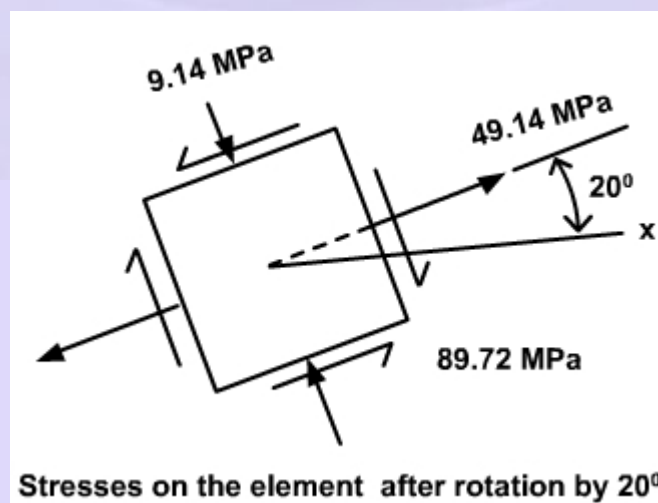
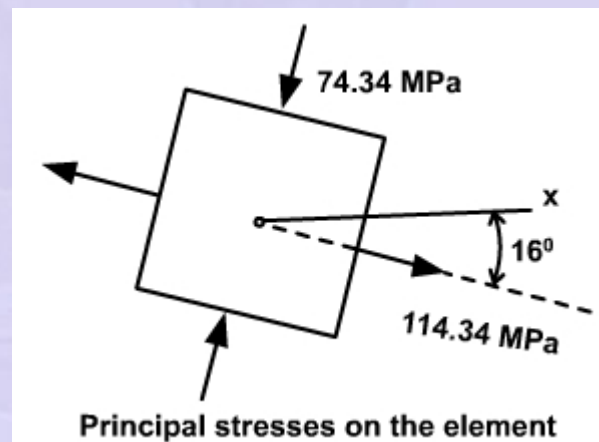
$$\begin{aligned}\text{Maximum shear stress} &= \pm \left[ \frac{114.34 - (-74.34)}{2} \right] \\ &= \pm 94.34 \text{ MPa}\end{aligned}$$

Normal stresses on the element after rotation by  $20^\circ$  counterclockwise

$$\begin{aligned}\sigma_{x'x'} &= \frac{100 + (-60)}{2} + \frac{100 - (-60)}{2} \cos(2 \times 20^\circ) + (-50) \sin(2 \times 20^\circ) \\ &= 49.14 \text{ MPa}\end{aligned}$$

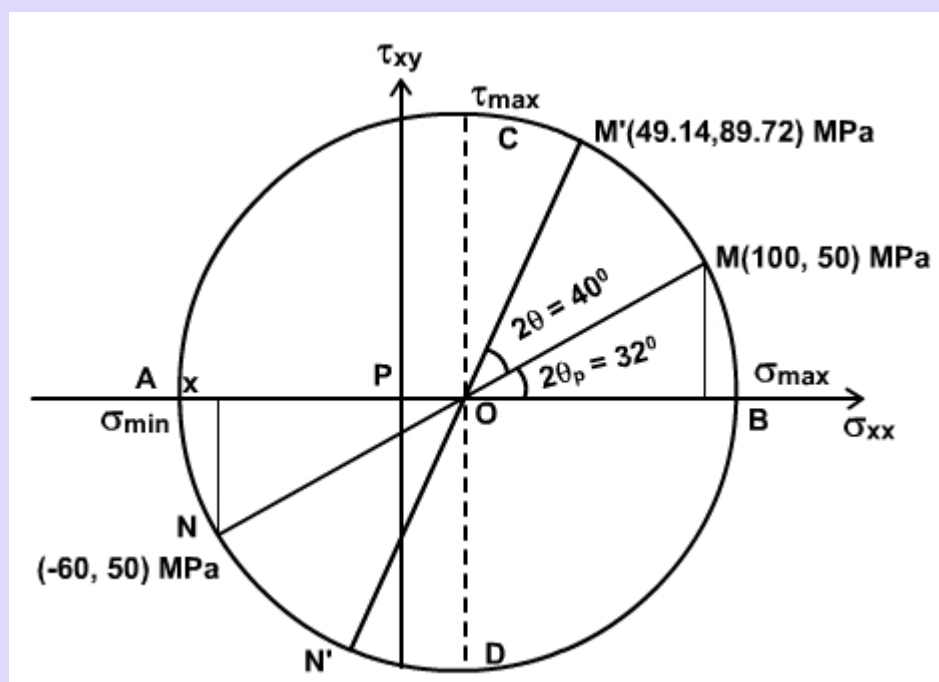
$$\begin{aligned}\sigma_{y'y'} &= \frac{100 + (-60)}{2} + \frac{100 - (-60)}{2} \cos(2 \times 110^\circ) + (-50) \sin(2 \times 110^\circ) \\ &= -9.14 \text{ MPa}\end{aligned}$$

$$\begin{aligned}\tau_{x'y'} &= - \left[ \frac{100 - (-60)}{2} \right] \sin(40^\circ) - 50 \cos(40^\circ) \\ &= -89.72 \text{ MPa}\end{aligned}$$



## II. Graphical Method:

Using the procedure discussed in the previous section, Mohr's circle is constructed as below.



$$\sigma_{max} = PB = 114.34 \text{ MPa}$$

$$\sigma_{min} = PA = -74.34 \text{ MPa}$$

$$\tau_{max} = OC \text{ or } OD = \pm 94.34 \text{ MPa}$$

Stresses on the element after rotating by  $20^\circ$  counterclockwise,

$$\sigma_{x'x'} = \text{abscissa of } PM' = 49.14 \text{ MPa}$$

$$\sigma_{y'y'} = \text{abscissa of } PN' = -9.14 \text{ MPa}$$

$$\tau_{x'y'} = \text{ordinate } PM' = 89.72 \text{ MPa}$$

**Top**